Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/18/2007
PI Name:	Menaker, Michael Ph.D.		
Project Title:	A Model of Circadian Disruption in the Space Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22904-4328	**Congressional District:**	5
Comments:	Deceased as of February 2021		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	15	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Yoshikawa, Tomoko (University of Virginia)		
Grant/Contract No.:	NCC 9-58-HPF00406		
Performance Goal No.:			
Performance Goal Text:			
	(1) ORIGINAL AIMS OF PROJECT SPECIFIC AIM I: Using rhythms of gene expression reported by a luciferase transgene, we will identify the signals linking brain and peripheral oscillators. SPECIFIC AIM II: We will evaluate potential countermeasures by exposing our experimental animals to conditions that disrupt the normal circadian phase map and then determining if the potential countermeasures are able to restore it rapidly. SPECIFIC AIM III: Investigate the effects of chronic low dose methamphetamine in synchronizing disrupted rhythms and evaluate related drugs as potential countermeasures. (2) THE KEY FINDINGS OF THE PROJECT We have extended our studies of the methamphetamine-sensitive circadian oscillator (MASCO) and have discovered a sex difference in the effect of methamphetamine on C3H mice, i.e., females are much less responsive than are males.		

Task Description:	We have initiated experiments to determine if estrogen is protective, and early results suggest that it is. We have explored the effects of methamphetamine on the internal circadian organization of mice bearing lesions of the suprachiasmatic nucleus (SCN). Untreated mice with such lesions have seriously disrupted phase relationships among internal organs. Treatment with chronic low levels of methamphetamine unexpectedly resynchronizes these internal phase relationships in addition to restoring behavioral rhythmicity, with possible benefits to integrated function. We are currently testing the effects of chronic methamphetamine on rhythmicity disrupted by constant light. These surprising results have led us to ask whether programmed feeding, which may act on the same circadian oscillator as methamphetamine, might also restore internal synchrony to arrhythmic mice. Preliminary results suggest that it does so. This is particularly exciting because programming meal times is a non-invasive and potentially beneficial way to treat circadian disorganization produced by the space environment. We have finished our studies of the phase-controlling signals to two internal organs, the ovary and the submaxillary gland, and have submitted two papers for publication. To summarize these findings: the ovary responds to pituitary hormones but not to neural signals; the salivary gland responds to both neural signals and to feeding time, but only if the neural inputs are cut. This variety of responses suggest that internal circadian organization is normally maintained by many different signals that are specific to particular organs or groups of organs and may be hierarchically organized. Understanding and eventually controlling them will be an important but difficult undertaking. (3) THE IMPACT OF THESE FINDINGS ON THE HYPOTHESES, OBJECTIVES AND SPECIFIC AIMS OF THE ORIGINAL PROPOSAL Our project is intended to provide some of the basic data that will be needed to minimize the effects of disrupting the circadian rhythms of astronauts and ground-based support personnel during their missions. Such disruption is an inevitable result of the work schedules that these men and women are required to follow. Disruption of circadian organization, which includes but is not limited to sleep deprivation, has repeatedly been shown to affect alertness, cognition, and performance on a variety of tasks. In order to minimize these effects, we have to understand the organization of the system that is being affected. In particular, we have to identify general signals such as meal timing that are capable of maintaining internal synchrony with minimal associated side effects. (4) THE PROPOSED RESEARCH PLAN FOR THE COMING YEAR During the coming year we will: • Expand our study of sex difference in response to methamphetamine. • Extend and confirm our initial results on the effects of methamphetamine and programmed feeding on disrupted circadian systems. • Use programmed meals as countermeasures under a variety of circumstances that parallel the conditions to which astronauts are exposed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The greatest potential benefit to American society is that the knowledge gained by this research will help both private industry and government organizations to design rational shift work schedules. It will also benefit astronauts and ground personnel subject to work schedules that disrupt circadian rhythmicity, and the basic science community working to understand mammalian circadian organization.
Task Progress:	• We have discovered a sex difference in the effects of methamphetamine on circadian rhythmicity. Females are less affected and appear to be protected by estrogen. • We have identified a hierarchy of circadian phase-shifting signals that influence the rhythmicity in salivary glands. Sympathetic neural inputs are dominant, but can be replaced by feeding schedules if the sympathetic nerves are cut. • We have shown that the circadian phase of the ovary does not respond to neural signals but does respond to pituitary hormones (LH and FSH). We hypothesize that the circadian oscillator in the ovary influences the timing of ovulation. This is a reasonable hypothesis given our findings, but runs counter to the strongly held prevailing idea that the timing of ovulation is exclusively controlled by the pituitary. • We have shown that disrupted internal circadian organization can be restored by chronic application of low dose methamphetamine and, more importantly, by rigid timing of meals.
Bibliography Type:	Description: (Last Updated: 03/07/2011)
Articles in Peer-reviewed Journals	Davidson AJ, Yamazaki S, Arble DM, Menaker M, Block GD. "Resetting of central and peripheral circadian oscillators in aged rats." Neurobiol Aging. 2006 Nov 25; [Epub ahead of print] PMID: 17129640 , Nov-2006
Articles in Peer-reviewed Journals	Davidson AJ, Sellix MT, Daniel J, Yamazaki S, Menaker M, Block GD. "Chronic jet-lag increases mortality in aged mice." Curr Biol. 2006 Nov 7;16(21):R914-6. PMID: 17084685 , Nov-2006
Articles in Peer-reviewed Journals	Yoshikawa T, Papadimas E, Cha S, Deng A, Pezuk P, Menaker M. "The ovary may contribute to the circadian timing of ovulation." Proc Natl Acad Sci U S A. In press, April 2007. , Apr-2007
Awards	Menaker M. "Harvard Medical School 2007 Peter C. Farrell Prize in Sleep Medicine. June 2007." Jun-2007